

Methods for assessing replicability of null results

There are both frequentist and Bayesian methods that can be used for assessing evidence for the absence of an effect. *Anderson and Maxwell (2016)* provide an excellent summary in the context of replication studies in psychology. We now briefly discuss two possible approaches – frequentist equivalence testing and Bayesian hypothesis testing – and their application to the RPCB, EPRP, and RPP data.

Frequentist equivalence testing

Equivalence testing was developed in the context of clinical trials to assess whether a new treatment – typically cheaper or with fewer side effects than the established treatment – is practically equivalent to the established treatment (*Wellek, 2010; Lakens, 2017*). The method can also be used to assess whether an effect is practically equivalent to an absent effect, usually zero. Using equivalence testing as a way to put non-significant results into context has been suggested by several authors (*Hauck and Anderson, 1986; Campbell and Gustafson, 2018*). The main challenge is to specify the margin $\Delta > 0$ that defines an equivalence range $[-\Delta, +\Delta]$ in which an effect is considered as absent for practical purposes. The goal is then to reject the null hypothesis that the true effect is outside the equivalence range. This is in contrast to the usual null hypotheses of superiority tests which state that the effect is zero or smaller than zero, see Figure 2 for an illustration.

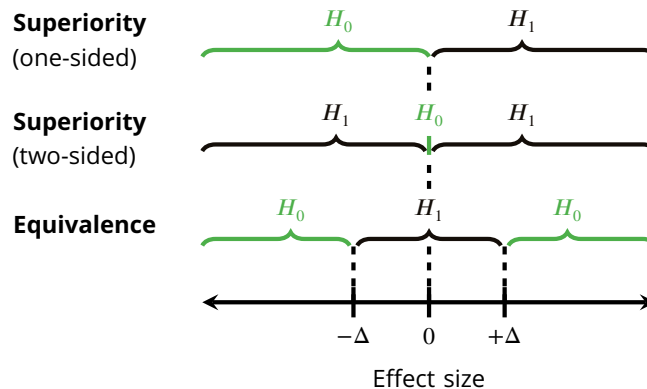


Figure 2. Null hypothesis (H_0) and alternative hypothesis (H_1) for superiority and equivalence tests (with equivalence margin $\Delta > 0$).

To ensure that the null hypothesis is falsely rejected at most $\alpha \times 100\%$ of the time, the standard approach is to declare equivalence if the $(1 - 2\alpha) \times 100\%$ confidence interval for the effect is contained within the equivalence range, for example, a 90% confidence interval for $\alpha = 0.05$ (*Westlake, 1972*). This procedure is equivalent to declaring equivalence when two one-sided tests (TOST) for the null hypotheses of the effect being greater/smaller than $+\Delta$ and $-\Delta$, are both significant at level α (*Schirmann, 1987*). A quantitative measure of evidence for the absence of an effect is then given by the maximum of the two one-sided p -values – the TOST p -value (*Greenland, 2023*, section 4.4). In case a dichotomous replication success criterion for null results is desired, it is natural to require that both the original and the replication TOST p -values are smaller than some level α (conventionally $\alpha = 0.05$). Equivalently, the criterion would require the $(1 - 2\alpha) \times 100\%$ confidence intervals of the original and the replication to be included in the equivalence region. In contrast to the non-significance criterion, this criterion controls the error of falsely claiming replication success at level α^2 when there is a true effect outside the equivalence margin, thus complementing the usual two-trials rule in drug regulation (*Senn, 2021*, Section 12.2.8).

Returning to the RPCB data, Figure 3 shows the standardized mean difference effect estimates

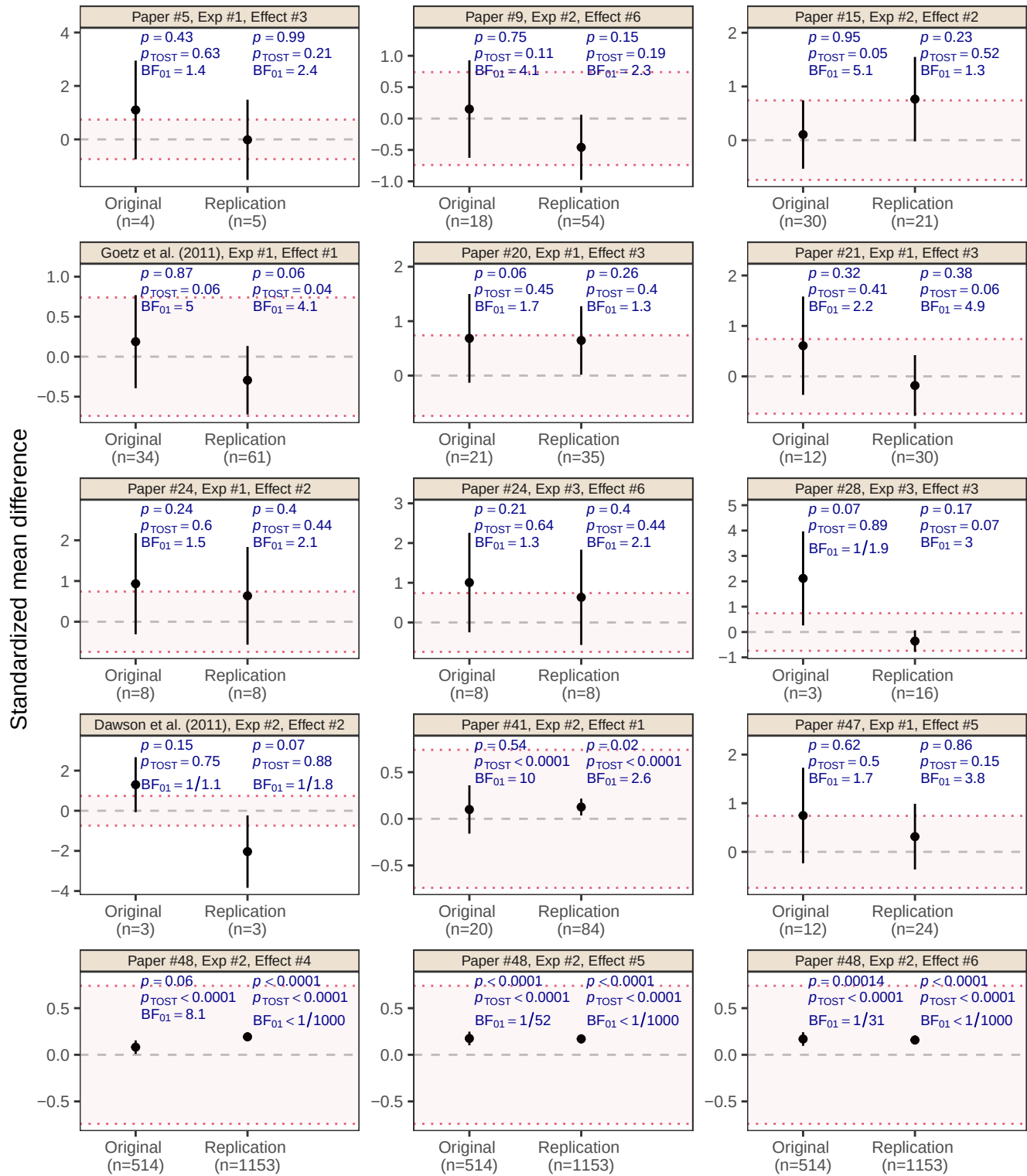


Figure 3. Effect estimates on standardized mean difference (SMD) scale with 90% confidence interval for the 15 “null results” and their replication studies from the Reproducibility Project: Cancer Biology (Errington et al., 2021). The title above each plot indicates the original paper, experiment and effect numbers. Two original effect estimates from original paper 48 were statistically significant at $p < 0.05$, but were interpreted as null results by the original authors and therefore treated as null results by the RPCB. The two examples from Figure 1 are indicated in the plot titles. The dashed gray line represents the value of no effect (SMD = 0), while the dotted red lines represent the equivalence range with a margin of $\Delta = 0.74$, classified as “liberal” by [Wellek \(2010, Table 1.1\)](#). The p -value p_{TOST} is the maximum of the two one-sided p -values for the null hypotheses of the effect being greater/less than $+\Delta$ and $-\Delta$, respectively. The Bayes factor BF_{01} quantifies the evidence for the null hypothesis H_0 : SMD = 0 against the alternative H_1 : SMD $\neq$ 0 with normal unit-information prior assigned to the SMD under H_1 .

with 90% confidence intervals for all 15 effects which were treated as null results by the RPCB.¹ Most of them showed non-significant p -values ($p > 0.05$) in the original study. It is noteworthy, however, that two effects from the second experiment of the original paper 48 were regarded as null results despite their statistical significance. According to the non-significance criterion (requiring $p > 0.05$ in original and replication study), there are 11 “successes” out of total 15 null effects, as reported in Table 1 from *Errington et al. (2021)*.

We will now apply equivalence testing to the RPCB data. The dotted red lines in Figure 3 represent an equivalence range for the margin $\Delta = 0.74$, which *Wellek (2010, Table 1.1)* classifies as “liberal”. However, even with this generous margin, only 4 of the 15 study pairs are able to establish replication success at the 5% level, in the sense that both the original and the replication 90% confidence interval fall within the equivalence range (or, equivalently, that their TOST p -values are smaller than 0.05). For the remaining 11 studies, the situation remains inconclusive and there is no evidence for the absence or the presence of the effect. For instance, the previously discussed example from *Goetz et al. (2011)* marginally fails the criterion ($p_{\text{TOST}} = 0.06$ in the original study and $p_{\text{TOST}} = 0.04$ in the replication), while the example from *Dawson et al. (2011)* is a clearer failure ($p_{\text{TOST}} = 0.75$ in the original study and $p_{\text{TOST}} = 0.88$ in the replication) as both effect estimates even lie outside the equivalence margin.

The post-hoc specification of equivalence margins is controversial. Ideally, the margin should be specified on a case-by-case basis in a pre-registered protocol before the studies are conducted by researchers familiar with the subject matter. In the social and medical sciences, the conventions of *Cohen (1992)* are typically used to classify SMD effect sizes (SMD = 0.2 small, SMD = 0.5 medium, SMD = 0.8 large). While effect sizes are typically larger in preclinical research, it seems unrealistic to specify margins larger than 1 on SMD scale to represent effect sizes that are absent for practical purposes. It could also be argued that the chosen margin $\Delta = 0.74$ is too lax compared to margins commonly used in clinical research (*Lange and Freitag, 2005*). We therefore report a sensitivity analysis regarding the choice of the margin in Figure 4 in Appendix A. This analysis shows that for realistic margins between 0 and 1, the proportion of replication successes remains below 50% for the conventional $\alpha = 0.05$ level. To achieve a success rate of $11/15 = 73\%$, as was achieved with the non-significance criterion from the RPCB, unrealistic margins of $\Delta > 2$ are required.

Appendix B shows similar equivalence test analyses for the four study pairs with original null results from the RPP and EPRP. Three study pair results turn out to be inconclusive due to the large uncertainty around their effect estimates.

Bayesian hypothesis testing

The distinction between absence of evidence and evidence of absence is naturally built into the Bayesian approach to hypothesis testing. A central measure of evidence is the Bayes factor (*Kass and Raftery, 1995; Goodman, 1999; Dienes, 2014; Keyzers et al., 2020*), which is the updating factor of the prior odds to the posterior odds of the null hypothesis H_0 versus the alternative hypothesis H_1

$$\underbrace{\frac{\Pr(H_0 \text{ given data})}{\Pr(H_1 \text{ given data})}}_{\text{Posterior odds}} = \underbrace{\frac{\Pr(H_0)}{\Pr(H_1)}}_{\text{Prior odds}} \times \underbrace{\frac{\Pr(\text{data given } H_0)}{\Pr(\text{data given } H_1)}}_{\text{Bayes factor } \text{BF}_{01}}$$

The Bayes factor BF_{01} quantifies how much the observed data have increased or decreased the probability $\Pr(H_0)$ of the null hypothesis relative to the probability $\Pr(H_1)$ of the alternative. As such, Bayes factor are direct measures of evidence for the null hypothesis, in contrast to p -values, which are only indirect measures of evidence as they are computed under the assumption that the

¹There are four original studies with null effects for which two or three “internal” replication studies were conducted, leading in total to 20 replications of null effects. As done in the RPCB main analysis (*Errington et al., 2021*), we aggregated their SMD estimates into a single SMD estimate with fixed-effect meta-analysis and recomputed the replication p -value based on a normal approximation. For the original studies and the single replication studies we report the SMD estimates and p -values as provided by the RPCB.